Jonathan Poteet

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience developing and integrating production-grade enterprise applications. Experienced in TypeScript, Python, and Java, with a focus on distributed systems, software integration, application architecture, and CI/CD automation. Key technical contributor to GM's enterprise platform modernization, integrating services, APIs, and data across applications while contributing to micro-frontend architecture, system reliability, and secure software development.

TECHNICAL SKILLS

Frontend and Web: Angular, Solid.js, React, Next.js, TypeScript, JavaScript, HTML, CSS/SCSS

Backend and APIs: Node.js, Spring Boot, Kafka, Java, Python, REST APIs, SQL, jOOQ

Databases: MongoDB, MySQL, PostgreSQL

Cloud, DevOps, and Tools: Kubernetes, Docker, Podman, Ansible, Azure DevOps, CI/CD, Insomnia, Jira

Testing and Security: Vitest, automated testing, GitHub Advanced Security (GHAS), dependency management

Operations: Troubleshooting, production support, incident investigation

Systems and Networking: Linux, WSL, TCP/IP, Godot

PROFESSIONAL EXPERIENCE

Pod Juggler LLC

September 2026 — Present

Owner and Software Developer — Remote

- Formed a software consulting LLC focused on individual development projects and research-driven technical work; established the company foundation while preparing for client engagements.
- Defined service offerings, project scoping, and delivery processes for custom software development and research-based technical problem solving.
- Refined the company plan and market positioning to align prospective offerings with client needs and emerging opportunities.

General Motors

March 2021 — December 2025

Software Engineer — Roswell, GA

- Owned the My Orders frontend experience from checkout through delivery across four European markets, using Next.js, React, TypeScript, and Java and integrating Strapi CMS and enterprise APIs for localized vehicle and financial data.
- Collaborated with cross-functional partners to gather and clarify requirements for modernization of the enterprise technician web platform, then contributed to re-engineering legacy architecture with Angular, TypeScript, and Java to improve scalability, maintainability, and responsiveness.
- Built Docker and Podman container environments with WSL for multi-repository Linux microservices, achieving 100% environment parity and reducing deployment errors.
- Developed REST APIs and backend services with Java and Spring Boot; managed Maven dependencies and used SQL, jOOQ, and PostgreSQL for data access and containerized database management.
- Managed Azure DevOps CI/CD pipelines, improved API discoverability with Spring Boot Swagger, and automated software delivery to reduce manual deployment effort.
- Led migration of the My Orders platform to Solid.js in the DASH monorepo; introduced Vitest automated testing and Angular and ESLint strict modes to strengthen code quality.
- Remediated GitHub Advanced Security findings, managed third-party dependencies, and monitored Kubernetes pods supporting microfrontend services.

SELECTED PROJECTS

Sinkhole — Procedural Dungeon Crawler (Godot/C#/.NET)

Jan 2026 – Present

- **Infinite World Architecture:** Engineered a chunk-based streaming system in Godot 4 (C#/.NET) that generates world data at runtime using seeded random algorithms.
- **Procedural Generation:** Developed custom layout logic to construct non-linear, endlessly descending floors, balancing randomness with navigable structural constraints.
- **Performance Optimization:** Decoupled generation logic from the main render thread to prevent frame-rate stutters during high-intensity procedural generation cycles.

World State AI — Resource Optimization Simulation (Python/LISP)

Aug 2025 – Oct 2025

- **Expected Utility Algorithm:** Engineered an incremental algorithm to evaluate and rank action schedules based on calculated expected utility and reward functions.
- **Decision Logic:** Implemented a parser to translate complex state transitions into executable schedules, identifying optimal resource trades to maximize global state stability.
- **Scalable Modeling:** Developed weights for diverse resource types, enabling the simulation of complex international trade and transformation resource loops.

Probabilistic Sensor Verification (Python/PRISM)

Oct 2025 – Nov 2025

- **Formal Verification & Modeling:** Developed a framework to evaluate sensor reliability using PRISM model checker, modeling gas/thermal sensors as DTMC to simulate failure states.
- **Safety & Risk Assessment:** Authored PCTL safety properties to analyze system reachability and hazard thresholds, identifying theoretical failure points and safety boundaries.
- **Statistical Analysis:** Conducted cross-validation of formal results against Monte Carlo simulations using StormPy, evaluating variance between mathematical methods and empirical simulation.

Kubernetes Cluster for Distributed Systems (K8s/Python)

Oct 2024 – Nov 2024

- **Cloud Orchestration:** Architected a K8s cluster across four VMs on Chameleon Cloud. Resolved complex node-to-node networking and resource contention challenges.
- **Distributed Systems:** Integrated Apache Kafka for high-throughput messaging, optimizing resource provisioning to scale throughput by 10x (up to 5,000 messages per node).
- **Automation:** Authored custom Ansible playbooks and designed Dockerfiles to ensure environment parity and enforced deployment immutability via CI/CD.

PROFESSIONAL PORTFOLIO — JONATHANPOTEET.COM

- **React + Next.js:** Built with React and Next.js and deployed via GitHub Pages to showcase modern frontend architecture, component-driven development, and scalable web experiences.
- **Creative Engineering:** Combines formal software engineering with procedural generation and 3D design (Three.js, Blender) to build interactive, logic-driven environments.
- **Design-to-Code Execution:** Translates visual design concepts into polished frontend experiences with careful attention to structure, interaction, and implementation quality.

EDUCATION

Vanderbilt University — December 2025

Master of Science in Computer Science — GPA: 3.97

Florida State University — December 2020

Bachelor of Science in Information Technology — Minor in Business — GPA: 3.80

CERTIFICATIONS

CompTIA Security+ (SY0-701) — 2026

Global Citizenship Certificate — Florida State University — 2020

CAMPUS AND COMMUNITY INVOLVEMENT

Study Abroad: FSU International Programs — London, England — Summer 2019

Air Force ROTC (AFROTC) — Detachment 145 — 2017 — 2018

Florida American Legion Boys State — Delegate — 2016

Eagle Scout, Boy Scouts of America — 2016